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Your trip, your way: An adaptive tourism recommendation system

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HIGHLIGHTS

- Offering insights into improving tourism recommendation system in uncertain context.
- Incorporating tourists' changing on-site behavior in designing tour itineraries.
- A hybrid optimization structure is used to balance the efficiency and performance.
- Proposing a dynamic mechanism to enhance the system's flexibility and adaptability.

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ABSTRACT

Considerable changes in the behavior of tourists have been observed because the call for their customized experiences prevails in the market. In addition, advancements in artificial intelligence have significantly promoted the development of tourism recommendation systems (TRS), which substantially contributes in enhancing tourists' personalized experiences. However, existing systems are facing increasing challenges in handling uncertain environments and the changing on-site behavior of tourists. This study intends to address these challenges by proposing an adaptive TRS, which mainly consists of three modules: data collection, itinerary design, and itinerary adjustment. To evaluate the proposed system's performance, a case study was conducted in Xiamen, China. Results indicated that our system can adapt to uncertain environments and tourists' changing on-site behavior. Tourism organizations could integrate our system in upgrading their information service platforms as part of their smart tourism construction to enhance the competitiveness of destinations.

1. Introduction

Considerable changes in the behavior of tourists have been observed in the postmodern era because the call for tourists' customized experiences prevails in the market [1,2]. In recent decades, the rapid development of artificial intelligence (AI) and related technologies has also led to the widespread application of intelligent technology in the tourism industry [3]. Following these scenarios, the smart tourism recommendation system (TRS) has emerged as a field that warrants attention in tourism research [4], given its important role in enhancing the competitiveness of tourism destinations and improving tourist experiences, which has been increasingly recognized [5]. Compared with the recommendation system widely used in e-commerce, TRS is more complex because it needs to provide tourists with personalized itineraries and involves several interactive elements, various constraints, and conflicting objectives [6].

Nevertheless, although some progress has been made recently, several aspects of the TRS research remains to be further improved. First, tourist preferences significantly affect the selection of POIs (Points of Interest) and then influence the design of the itinerary. Most of the existing TRSs often require tourists to input their preferences for each POI in advance, which is markedly cumbersome, especially for tourist destinations with numerous POIs. Moreover, fully knowing all POIs in advance is difficult for most tourists, leading them to input inaccurate preference value. Second, most TRSs design personalized itineraries for tourists based on the assumptions of certainty and invariable, which are usually impractical. In practice, several unforeseeable factors, such as traffic uncertainty and waiting time, may force tourists to deviate from the designed itineraries. Furthermore, tourists often adjust their itineraries by themselves during their trip. For example, tourists may visit POIs that are not planned or cancel planned ones; their stay time in some POIs may be different from that planned. Therefore, uncertain

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environments and tourists' changing behavior should be considered when designing TRSs.

To fill in the aforementioned gaps, this study presents an adaptive TRS, which contains three modules, namely, data collection, itinerary design, and itinerary adjustment. Our TRS differs from existing systems in three main aspects. (1) An innovative approach of tourist profile learning is adopted to accurately collect tourist preferences. (2) A hybrid optimization structure is used to balance the efficiency and performance of itinerary design. (3) Lastly, a dynamic mechanism is proposed to enhance the flexibility and adaptability of the system. To confirm the system's performance, a case study was conducted in Kulangsu islet, southwest of Xiamen in China.

This study provides pertinent methodological and practical contributions to the TRS field. From the methodological perspective, we develop an adaptive system for providing tourists with personalized itineraries that can adapt to uncertain environments and their changing behavior. In terms of practical perspective, our system provides new perspectives for tourism managers to improve their personalized services, which are significant to their tourists.

The remainder of this paper is organized as follows. Section 2 reviews the related studies on tourists' on-site behavior and tourist trip design problem (TTDP). Section 3 illustrates in detail the proposed adaptive TRS. Section 4 describes the case study conducted to examine the performance of our system. Lastly, Section 5 provides the conclusions and proposes several suggestions for future research directions.

2. Literature review

2.1. On-site tourist behavior

The study of tourist behavior has benefited from its extended nature, incorporating ideas and concepts from various disciplines, such as geography, psychology, and sociology, leading to this field's rapid development [7]. Research from the psychological perspective has focused on the analysis of the motivations of tourist behavior from a micro perspective [8]. Research based on a geographic perspective has centered on the portrayal of tourist flows [9] and the spatial behavior of tourists [10]. Moreover, the contribution of sociology to the study of tourist behavior is to advance the study of tourist place attachment [11], tourist experience [12], and tourist host–guest relationships [13]. Tourist behavior, following the broadest psychological definition, implies the collection of observable behaviors and thought processes that originate and guide social life [14]. Accordingly, the aim of tourist behavior research is to acquire knowledge that can explain the explicit behavior and decision-making processes of tourists. Therefore, understanding the psychological changes in the decision-making process of tourist behavior is important for acquiring insights into tourist behavior.

Given that tourism itineraries can be chronologically divided into pre-site, on-site, and post-site [15], tourist behavior is an aggregated behavior that mainly includes pre-site motivation and destination selection behavior [8], on-site experience, and post-site willingness to revisit and recommend [16,17]. On-site trip, compared with pre- and post-trips, takes place in unconventional environments, where tourists need to continuously collect information, process information, and make decisions in the entire process, which is a typical dynamic decision-making process based on information processing theory [18]. In this process, tourists' behavior will be affected by their own and external environments, such that their behavior is in a state of continuous change; this fact makes the study on tourists' on-site behavior markedly complicated [19,20].

Tourists' on-site behavior can be affected by socioeconomic status, disposable income, education level, and other factors related to individuals [21], travel experience [22], cultural distance [23], and destination image bias [24]. Moreover, tourists tend to seek different experiences [25] or change their preferences [26] during the different phases of their trip, further changing their on-site behavior. In addition to tourists' own factors, external environmental factors also change their on-site behavior, such as weather [27], traffic condition [28], congestion of POIs [29], discount [30], online review [31], and some unexpected circumstances [32]. Spatial factors, such as distance between POIs [33], hotel location, and landform obstacles [34], also affect tourists' on-site behavior.

2.2. Tourist trip design problem

In the postmodern era, tourist behaviors significantly changed because the call for customized experiences has grown and currently prevails in the market [1,2]. Thus, designing customized itineraries has emerged as a significant research topic in the tourism field, facilitating tourists' understanding of the culture and social condition of the tourism destinations they are visiting within a certain time [5,35].

2.2.1. TTDP-related studies in different contexts

Existing TTDP-related studies can be dichotomized according to whether uncertainty is considered. Most studies have modeled TTDP under certainty contexts and incorporated various factors to make the designed routes approach real tourism situations [36,37]. For example, the time window [38,39], hotels/restaurants selection [40,41], transport mode selection [42], esthetic fatigue [43], carbon emission [44], POIs' spatial structure [45,46], and heterogeneous tourist groups [47–49]. However, all aforementioned studies have addressed TTDP under deterministic assumptions (i.e., various variables do not change over time).

Such variables as travel and service times may be stochastic owing to the effects of weather condition, traffic congestion, and unforeseen events [50], thereby converting the certainty of TTDP into uncertainty [51]. Liao and Zheng attempted to design personalized tour routes under the time-dependent stochastic environment [52]. In particular, travel time needed between POIs and wait time at a POI are treated as stochastic parameters to approximate considerably realistic settings. Karunakaran et al. set visit duration as a random variable and designed tour routes in such a stochastic condition [53]. In general, fitting distribution function of stochastic variables is a useful tool for handling uncertainty [54]. However, function fitting always needs extensive historical data, and the accuracy of function is also questionable [55]. Therefore, some studies have attempted to handle the uncertain TTDP under a fuzzy environment [56,57]. Matsuda et al. made the first contribution in the design of tour route by considering travel time as fuzzy [58]. Furthermore, some studies have developed tour routes considering profit and travel time constraints as fuzzy [54,59]. These studies have significantly improved the applicability and reliability of the designed tour routes.

The aforementioned studies have focused on how to maintain sufficient flexibility in an uncertain environment [52,54] or how to balance tourists' utility and the risk that trips cannot be completed smoothly in an uncertain environment [60]. In addition to uncertain environment, particularly prevalent for tourists is to adjust their itineraries by themselves. These facts necessitate considering the itineraries' flexibility and also real-time adjustments according to changes among tourists and environment. Therefore, the current study aims to present an adaptive TRS, which can adjust itineraries in real-time to adapt to the uncertain environment and the changing on-site behavior of tourists.

2.2.2. Different approaches for TTDP

The selection of solution approach remarkably affects the qualities of designed tour routes. Existing approaches can be divided into three categories: exact, heuristic, and metaheuristic approaches [61].

Exact approaches are generally applied to search the optimal solution for small-size instances with a small amount of vertices [62]. Yu et al. adopted linear programming to design tour route for a real scenario where only 20 nodes are considered [63]. Wang et al. used a branch and bound method to achieve optimal route design on instances of up to 58 POIs [64]. Moreover, such approaches as dynamic programming and branch and check have been introduced to handle TTDP [36,65]. However, TTDP is NP-hard [66], implying that metaheuristic or heuristic approaches are more effective in finding near-optimal solutions in rational calculation times [66]. In particular, heuristic algorithms may be grouped into constructive, improvement, and relational techniques [67]. Wang et al. used a constructive heuristic approach to deal with TTDP [64]. Maervoet et al. further improved the heuristic method on the basis of spatial filtering to arrange tour routes for tourists [68]. Gavalas et al. introduced a two-phases heuristic approach, which consists of POI clustering and route matching, to design tour routes [69]. Additional heuristic algorithms are found in the studies [70–72]. Metaheuristic algorithms have been most frequently used in relevant studies compared with the first two approaches. Metaheuristic algorithms are developed by guiding and modifying heuristic methods [6]. Such algorithms as guided local search [73], genetic algorithm (GA) [74], variable neighboring search (VNS) [75], and simulated annealing (SA) [76] have been introduced and improved to find approximate solutions for tour route designs. Moreover, some studies have integrated multiple metaheuristic algorithms to improve performance. Zheng et al. integrated GA and differential evolution algorithm (DEA) to mixed-variable optimization [43]. Ruiz-Meza et al. combined variable neighborhood descendent and greedy randomized search procedure to avoid local optimum and raise solution quality [57].

To design an adaptive TRS, real-time response speed, in addition to performance, is extremely important. Therefore, more effective optimization approaches and more efficient optimization structures are required to balance the relationship between system efficiency and performance.

3. Methodology

Our adaptive TRS mainly consists of three modules: data collection, itinerary design, and itinerary adjustment (as shown in Fig. 1). In the first module, the basic information of tourists and POIs is collected. In particular, tourists' preferences for each POI are also obtained based on feature matching. On the basis of the collected data, the itinerary design module designs itineraries for tourists to meet their needs under various constraints. In reality, tourists are likely not to follow a designed itinerary owing to changes in their mind or the environment. For this reason, the itinerary adjustment module is designed to improve the adaptability of the system. The gap between the actual and designed itineraries is specifically compared in real time based on the support of

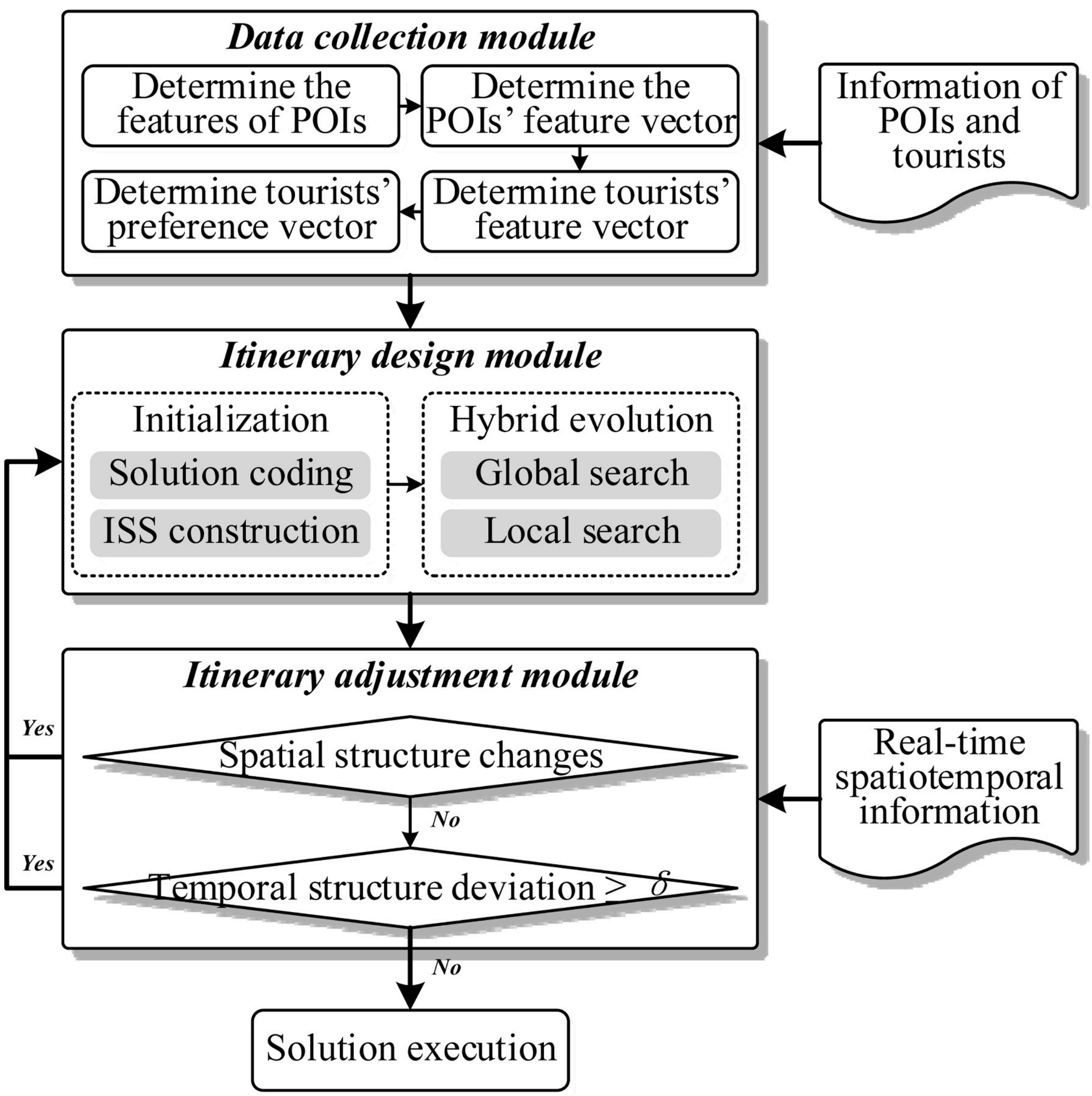

Fig. 1. Overall framework of the proposed system.

real-time positioning function (e.g., Global Positioning System). If the difference reaches a certain threshold, then the itinerary design module is restarted.

3.1. Data collection module

Data collection considerably affects an itinerary design; thus, the accurate collection of tourist and POI information is highly important. In this module, the basic information of POIs, such as their time windows, average duration time of previous tourists, and distance between POIs, is collected. In addition, tourists' time budget and must-play and must-avoid POIs should be obtained. The most important issue in this module is to gather tourists' preferences for each POI, which will affect the selection of POIs and stay time in each selected POIs and then affect the design of the entire itinerary. In most existing TRSs, tourist preference values are often obtained through questionnaires; that is, tourists are required to fill in the willingness to visit each POI before obtaining the personalized itinerary [41,60]. This task is highly cumbersome, especially for tourist destinations that include numerous POIs. Moreover, fully knowing all POIs in advance is difficult for most tourists, leading to the inaccurate preference values they input.

To fill in the aforementioned gap, we calculate tourists' preference for each POI through the feature matching of tourists and POIs. This method is widely used in content-based recommendation in the field of e-commerce. In this study, we collect tourists' preference values, combined with the characteristics of tourism context, according to the following four steps. (1) *Determine features that can fully depict POIs*. In recent years, several studies have proposed some features of POIs. Leask summarized seven categories of tourist destinations [77]. On this basis, Yeh and Cheng proposed 9 categories and 27 constructs [78]. Our study adopts eight features to depict POIs according to China's national standard of the "classification, investigation and evaluation of tourism resources" [79], namely, land scenery, water scenery, biological scenery, astronomical and climatic scenery, relics, architecture and facility, tourism commodity, and humanistic activity. (2) *Determine POIs' feature vector*. Academic experts, senior practitioners, and experienced tourists who are familiar with POIs to be evaluated are invited to estimate the extent to which the features contribute to the characteristics of POIs in the 0–1 range (1 represents the highest degree; 0, no at all). The feature vector for the ith POI can be denoted as $\boldsymbol{x}_i = [x_i^1, x_i^2, \ldots, x_i^K]$. (3) *Determine tourists' feature vector*. Tourists are required to fill in their preference for each POIs' features in the 0–1 range (0 means no interest in this feature; 1 denotes highest interest to the feature) to obtain their feature vector: $\boldsymbol{y}_j = [y_i^1, y_i^2, \ldots, y_i^K]$. (4) *Determine tourists' preference vector*. We calculate tourists' preference value for each POI by measuring the similarity between the feature vectors of POIs and tourists based on the COSINE similarity method, which is widely used in the field of recommendation. The similarity value reflects tourists' preference degree for POI, which can be calculated based on Eq. (3.1). The greater the value, the higher the degree of preference, and vice versa. In this equation, k is the number of features, $\boldsymbol{x}_i$ is the feature vector of the ith POI, and $\boldsymbol{y}_j$ represents the feature vector of the jth tourist.

$$sim(\boldsymbol{x}_i, \boldsymbol{y}_j) = \cos(\boldsymbol{x}_i, \boldsymbol{y}_j) = \frac{(\boldsymbol{x}_i, \boldsymbol{y}_j)}{\|\boldsymbol{x}_i\| \cdot \|\boldsymbol{y}_j\|} = \frac{\sum_{k=1}^{K} x_i^k \cdot y_j^k}{\sqrt{\sum_{k=1}^{K} (x_i^k)^2 \cdot \sum_{k=1}^{K} (y_j^k)^2}} \tag{3.1}$$

3.2. Itinerary design module

The itinerary design module, as the core module of our TRS, involves providing tourists with personalized trips and meeting their objectives with consideration of a multitude of constraints [80].

3.2.1. Mathematics model

This subsection describes the mathematical model. Table 1 presents the variables used in this study.

3.2.1.1. Objectives of the model. The objective of itinerary design module is to maximize total utility within the limits of a given time budget. Generally, tourists may specify a set of itinerary start locations ($\boldsymbol{V_S}$) and a set of itinerary end locations ($\boldsymbol{V_E}$). The tourist starts the itinerary from one of the start locations, and finishes the trip from one of the end locations. We ignore the duration of tourists' stay at the start and end locations and assume that they will receive no utility from such locations. Morever, we assume that tourists can visit each POI at most once and have no utility during transit or waiting time. Utility gained by tourists at each stage depends on the vertex they visited (Λ_j). In particular, utility is decided by tourists' preference for Λ_j and the time spent at Λ_j (d_j). As marginal subjective sensation ($MS_i(t)$) generally decreases as time spent at the same vertex increases, utility gained from each vertex is a decreasing function of time. Thus, the utility that tourists achieve at the jth stage can be calculated using Eq. (3.2), where $MS_i(t)$ is a non-negative decreasing function of time (i.e., the marginal subjective sensation of the tourist gained from v_i at moment t). In addition, x_{ij} is a 0–1 discrete variable: $x_{ij} = 1$ if tourists visit v_i at the jth stage, and 0 otherwise; t_j^s denotes tourists' actual start time of visiting Λ_j; and t_j^a represents the arrival time at Λ_j. Because the vertices' time windows may cause early tourists to wait, t_j^s and t_j^a are not equal in many cases. Thus, t_j^s can be calculated according to Eq. (3.3), where $[t_i^o, t_i^c]$ is the time window of v_i.

$$u_j = \int_{t_j^s}^{t_j^e} \left\{ \sum_{i=1}^{N} \left[MS_i(t) \cdot p_i \cdot x_{ij} \right] \right\} dt \tag{3.2}$$

$$t_j^s = \max\left[t_j^a, t_i^o\right] \tag{3.3}$$

The total utility that tourists achieved during their itinerary is described as Eq. (3.4), which is the objective function.

$$Max \quad U = \sum_{j=1}^{M} u_j \tag{3.4}$$

Table 1
The descriptions of parameters and variables.

Notation	Descriptions
$\boldsymbol{V_S}$	Set of itinerary start locations, $i = 1, 2, \ldots, N_1$
$\boldsymbol{V_{POI}}$	Set of points of interest (POIs), $i = 1, 2, \ldots, N_2$
$\boldsymbol{V_E}$	Set of itinerary end locations, $i = 1, 2, \ldots, N_3$
v_i	The ith vertex, $v_i \in \boldsymbol{V_S} \cap \boldsymbol{V_{POI}} \cap \boldsymbol{V_E}$. A vertex generally refers to a location, which can be the start-end location of an itinerary or a point of interest (POI)
$\boldsymbol{N}$	Number of vertices, $N = N_1 + N_2 + N_3$
t_i	Average time spent by previous tourists at v_i
$[t_i^o, t_i^c]$	Time window of v_i
$t(v_i, v_{i+1})$	Travel time needed between vertices (v_i, v_{i+1})
T	Budgeted time available to the tourist
τ	Time that the tourist plans to start the itinerary
p_i	Tourist's preference value for v_i, $p_i \in [0, 1]$
M	Total stages of the tourist
Λ_j	Vertex visited at the jth stage for the tourist
t_j^a	Tourist's time of arrival at vertex Λ_j
t_j^s	Tourist's actual start time of visiting vertex Λ_j
t_j^e	Tourist's time of departure from vertex Λ_j
u_j	Utility gained at the jth stage for the tourist
U	Total utility of the tourist gained during the itinerary
$\boldsymbol{S_C}$	Set of compulsory POIs
$\boldsymbol{S_A}$	Set of "must-avoid" POIs
y_{ij}	y_{ij} equals 1 if the tourist visits v_j right after v_i, and 0 otherwise
x_{ij}	$x_{ij} = 1$ if the tourist visits v_i at the jth stage; otherwise, $x_{ij} = 0$

3.2.1.2. Constraints of the model. In designing available itineraries, two types of constraints should be followed. (1) Permanent technical constraints guarantee that the designed itineraries are valid and have practical significance, as shown in Eqs. (3.5)–(3.9). (2) Personalized constraints represents tourists' specific preferences and requirements, as shown in Eqs. (3.11)–(3.12) [6]. In particular, Eqs. (3.5)–(3.6) ensures that the tourist should start the itinerary from one of the start locations at the first stage, and finish the trip from one of the end locations at the last stage. In addition, Eq. (3.7) restricts that from the second to the $M-1$th stages, only one POI is visited at each stage.

$$\sum_{v_i \in \boldsymbol{V_S}} x_{i0} = \sum_{v_k \in \boldsymbol{V_E}} x_{kM} = 1 \tag{3.5}$$

$$\sum_{j=1}^{M} \sum_{v_i \in \boldsymbol{V_S}} x_{ij} = \sum_{j=1}^{M} \sum_{v_i \in \boldsymbol{V_E}} x_{ij} = 1 \tag{3.6}$$

$$\sum_{v_i \in \boldsymbol{V_{POI}}} x_{ij} = 1, j = 2, 3, \ldots, M-1 \tag{3.7}$$

Eqs. (3.8) and (3.9) ensure the time and path connectivity, respectively, where y_{ij} is a 0–1 discrete variable: y_{ij} equals 1 if the tourist visits v_j right after v_i, and 0 otherwise.

$$\begin{cases} t_{j+1}^{a} = t_j^{e} + t(\Lambda_j, \Lambda_{j+1}) \\ t_0^{s} = \tau \end{cases} \tag{3.8}$$

$$\sum_{v_i \in \boldsymbol{V_S} \cup \boldsymbol{V_{POI}}} y_{ij} - \sum_{v_l \in \boldsymbol{V_{POI}} \cup \boldsymbol{V_E}} y_{jl} = 0, \quad \forall v_j \in \boldsymbol{V_{POI}}; v_i \neq v_j, v_j \neq v_l \tag{3.9}$$

On the basis of the assumption that tourists do not visit the same POI repeatedly, Eq. (3.10) restricts that each POI is visited, at most, once. Before starting their itineraries, tourists usually have a time budget and a list of "must-visit" or "must-avoid" POIs in mind. Tourists' experience will be severely affected if itineraries exclude mandatory POIs or include unwanted POIs. To prevent these situations, Eq. (3.11) guarantees that tourists' compulsory POIs are included in their itineraries, POIs to be avoided are excluded, where $\boldsymbol{S_C}$ and $\boldsymbol{S_A}$ denote the set of compulsory POIs and "must-avoid" POIs, respectively. Meanwhile, Eq. (3.12) ensures that the total time, including traffic, duration, and waiting times, in the destination is less than the time budget T, where t_M^a is the tourists' arrival time at the exits at the last stage.

$$\sum_{j=1}^{M} x_{ij} \leq 1, v_i \in \boldsymbol{V_{POI}} \tag{3.10}$$

$$\begin{cases} \sum_{j=1}^{M} x_{ij} = 1, v_i \in \boldsymbol{S_C} \\ \sum_{j=1}^{M} x_{ij} = 0, v_i \in \boldsymbol{S_A} \end{cases} \tag{3.11}$$

$$\tau + t_M^a \leq T \tag{3.12}$$

3.2.2. Solution approach

TTDP is an NP-hard problem; it is difficult to solve large-scale TTDP through exact algorithms. Therefore, a multitude of heuristic and metaheuristic algorithms have been proposed to solve the trip design problem in recent studies, such as GA and DEA [43,52]; ant colony algorithm (ACO) and DEA [81]; GA, VNS, and DEA [41]; and particle swarm optimization algorithm (PSO) and DEA [60]. These algorithms have good applicability in their own scenarios. For our system, efficiency (i.e., the ability to respond and adjust in real-time) is also highly important in addition to performance (i.e., the ability to design itineraries with as high utility as possible) because the system needs to be able to respond to itinerary changes as rapidly as possible [82]. Therefore, in view of the characteristics of the existing algorithms, the current study proposes a heuristic approach combining PSO and DEA in designing itineraries to balance the relationship between system efficiency and performance. In this module, PSO is applied to optimize the itinerary's spatial structure (i.e., POI selection and sequencing), and DEA is adopted to optimize the temporal structure (i.e., the time spent at POIs). We also use the hybrid optimization structure introduced in [52] to provide additional efficiencies for our system. Two steps are involved in the module, namely, (1) coding solution and construction of the initial solution set (**ISS**) and (2) hybrid optimization, which will be introduced as follows.

3.2.2.1. Solution coding and ISS construction. Most evolutionary algorithms need to determine the solution's dimension in advance [43], which is unsuitable for our problem because the number of POIs visited by tourists usually changes. In addition, the spatial structure (i.e., POI selection and sequencing) and temporal structure (i.e., the time spent at each selected POI) of itineraries need to be optimized simultaneously. Thus, a double-layer, variable-length chromosome proposed in [43] is used to code the itinerary (solution), in which the upper layer denotes the chosen POI and sequencing, while the lower layer indicates the temporal structure, that is, the average time spent by previous tourists at the corresponding vertex (as shown in Fig. 2). The performance of the approach is closely related to the quality of **ISS**, and two approaches are widely used to construct **ISS**, namely, random and greedy approaches [52]. To balance the relationship between the solutions' quality and diversity, the improved greedy algorithm introduced in [43] is applied to construct **ISS**.

3.2.2.2. Hybrid optimization. In the hybrid optimization structure, the spatial structure and temporal structure are optimized using PSO and DEA, respectively. Given its superior search performance [83], PSO is highly suitable for our problem. PSO is a swarm-based optimization algorithm motivated by animal behavior. The swarm conforms cooperatively to search for a solution, and each particle in the swarm keeps changing the search pattern guided by their own and the swarm's best solutions [84]. Eqs. (3.13) and (3.14) show these evolution processes, where z_j^t represents the solution of the jth particle obtained in the tth iteration, z_j^{t+1} denotes that in the $(t+1)$th iteration, and v_j^{t+1} is the velocity of moving from z_j^t to z_j^{t+1}, as shown in Eq. (3.14). In this equation, $Pbest_j^t$ denotes the best solution of the jth particle, $Gbest^t$ indicates the best solution of the entire swarm, ω is the inertia weight, σ_1 is the learning intensity with individual optimum, and σ_2 denotes learning intensity with global optimum. DEA is simple, has good convergence, and is ideally situated for continuous optimization problems [85]. Thus, DEA has been widely used to optimize a route's temporal structure in TTDP-related studies [60].

$$z_j^{t+1} = z_j^t + v_j^{t+1} \tag{3.13}$$

$$v_j^{t+1} = \omega v_j^t + \sigma_1 \left(Pbest_j^t - z_j^t\right) + \sigma_2 \left(Gbest^t - z_j^t\right) \tag{3.14}$$

The hybrid optimization structure introduced in [52] is used to further improve the efficiency of our adaptive TRS. Compared with the general evolutionary structure, the main difference of the hybrid optimization structure is that the optimization process is divided into global and local searches, and different optimization methods are selected in different types of search. Three main issues should be determined: (1) global or local search, (2) optimization objectives of the global and local searches; and (3) optimization method for each type of search. For the first issue, we determine the search type by evaluating the solution of two adjacent iterations. If the solution obtained in the current iteration (BFV_C) is significantly better than that of the previous iteration (BFV_P) (i.e., their ratio δ exceeds a certain threshold ξ), then the probability of finding the best solution near this solution space is substantial. In this situation, the local search is adopted; otherwise, the global search is

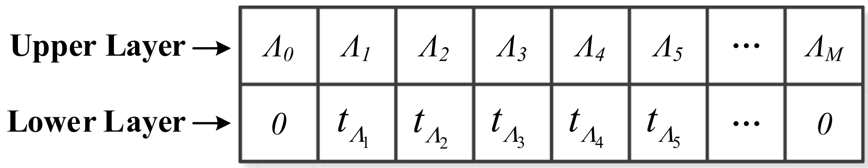

Fig. 2. Solution coding.

selected. For the second issue, given that the influence of spatial structure on the solution performance is significantly higher than that of the temporal structure, we regard spatial and temporal structure optimizations as the global and local searches, respectively. For the last issue, various methods are used in the global and local searches owing to the different objectives. Spatial structure optimization involves discrete variables. Thus, we integrate a high σ_1, low σ_2, and low-frequency DEA in the global search. Conversely, the temporal structure belongs to a continuous variable. Therefore, we integrate a low σ_1, high σ_2, and high-frequency DEA for local search. The hybrid evolution procedure is shown in Fig. 3.

3.3. Itinerary adjustment module

In view of uncertain environments and variations in tourists' on-site behavior, their actual itineraries may deviate from the designed ones. For example, some tourists may temporarily add POIs or cancel the planned POIs, which we call spatial structural change. In addition, the deviation of the temporal structure of itineraries often occurs. For example, tourists may stay longer or shorter than the planned time in POIs, and they may spend more or less time in traffic owing to changes in road traffic conditions. To improve the adaptability of the designed itineraries, our system incorporates the itinerary adjustment module to update the recommended itineraries according to tourists' actual itineraries. From the adaptability perspective, when tourists' actual itineraries deviate from the designed ones, the system should redesign the itineraries in time to meet the needs of tourists. However, the over-susceptibility of the system (i.e., frequent redesigning of itineraries) may cause excessive interference to tourists. Therefore, we need to design a mechanism to ensure the adaptability of the system while avoiding over-sensitivity.

We initially judge whether the spatial structure of the actual itinerary has changed compared with the designed itinerary (i.e., to judge whether tourists add unplanned POIs or delete planned POIs). If the spatial structure changes, then we restart the itinerary design module; otherwise, we judge whether the temporal structure of the itinerary changes. Changes in tourists' stay time at POIs and traffic time between POIs may cause changes in itineraries' temporal structure. However, to avoid system over-sensitivity, the itinerary design module will be restarted only when the change in the temporal structure exceeds a certain threshold. To do this, let t_i^a and $\widetilde{t}_i^a$ denote the planned arrival and actual arrival times, respectively, of POI_i; and t_i^d and $\widetilde{t}_i^d$ denote the planned departure and actual departure times, respectively, of POI_i. If $\left|\widetilde{t}_i^a - t_i^a\right| \geq \delta$ or $\left|\widetilde{t}_i^d - t_i^d\right| \geq \delta$ (i.e., the time deviation has exceeded the threshold δ), then the itinerary design module should be restarted. Evidently, system sensitivity generally depends on the value of the threshold (δ). Note that when restarting the itinerary design module, the starting location of the new itinerary is the current position. In addition, based on the assumption that tourists do not repeatedly visit the same POIs, already visited POIs will not appear in the following itinerary.

4. Performance evaluation

4.1. Case study area

To produce customized day itineraries to adapt to uncertain environments and tourists' changing behavior, deciding on the location for the case study should consider tourist destinations and tourism popularity of such destinations. Kulangsu (or Gulangyu), an islet with an area of 1.87 square kilometers located southwest of Xiamen City in China (Fig. 4), is selected for its reputation as a popular tourist destination and World Heritage Site. Kulangsu currently operates three ferries: Piano, Sanqiutian, and Neicuoao Ferry Terminals. Given that the first ferry is dedicated to citizens, tourists can only use the last two ferries (denoted as yellow nodes in Fig. 4). With Kulangsu's multicultural history, diverse architectural styles, and winding coastline, the city was host to over 13.3 million tourists in 2019. Among them, about 12 million tourists participated in day tours. Kulangsu's many POIs make self-initiated itineraries extremely challenging for tourists. TTDP is pertinent in improving tourist experiences and ensuring the tourism competitiveness of Kulangsu. Fig. 4 shows the map of Kulangsu and the spatial distribution of its 44 main POIs (denoted as red nodes) and 2 ferry terminals.

4.2. Data collection

4.2.1. Basic information of POIs

Among Kulangsu's numerous POIs, this study only considers the 44 main POIs according to their popularity and rankings in travel websites. The locations of the 44 POIs are presented in Fig. 4, and the serial numbers, names, and time windows of the POIs are listed in the first three columns of Table 2. The time spent by historical tourists at each POI (t_i) profoundly affects the construction initial solution. Data on t_i presented in Column 4 of Table 2 were gathered by combining the survey results of tourists and staff of Kulangsu. Moreover, determining the feature vectors of POI is extremely important because they significantly affect the preference value of tourists. To gather the feature vector of each POI, three groups of experts with different backgrounds were interviewed to guarantee that the results are representative and comprehensive [78]: (a) industry group: senior managers and tour guides of Kulangsu; (b) academic group: experienced scholars who know Kulangsu extremely well; and (c) tourist group: experienced tourists who have visited Kulangsu numerous times and are familiar with the 44 POIs. Each expert filled in data on a scale from 0 to 1 to evaluate the extent to which features contributed to the characteristics of POIs (1 indicating the highest, and 0 indicating none at all). We integrated the experts' scores and obtained the feature vector for each POI, as shown in Column 5 of Table 2.

4.2.2. Basic information of tourists

Tourist data were collected at Sanqiutian Ferry (v_{45}) and Neicuoao Ferry (v_{46}) to ensure the representativeness of the sample. The investigators were stationed at the exits of the ferries and invited passengers out of the ferries to participate in the survey. Through a brief oral interview, the purpose of the passengers' visit to Kulangsu and tour option (day tour or multi-day tour) were determined. Only passengers who intended to travel in Kulangsu and chose day-tour option were invited to participate in further investigations. Pictures and descriptions related to the eight features were shown to the respondents. They were asked to independently report their preference for these features in the range of "0" (no interest in the feature) to "1" (strong interest in the feature), as shown in Column 2 of Table 3. Combined with POIs' feature shown in Table 3, tourists' preference vector for the 44 POIs can be

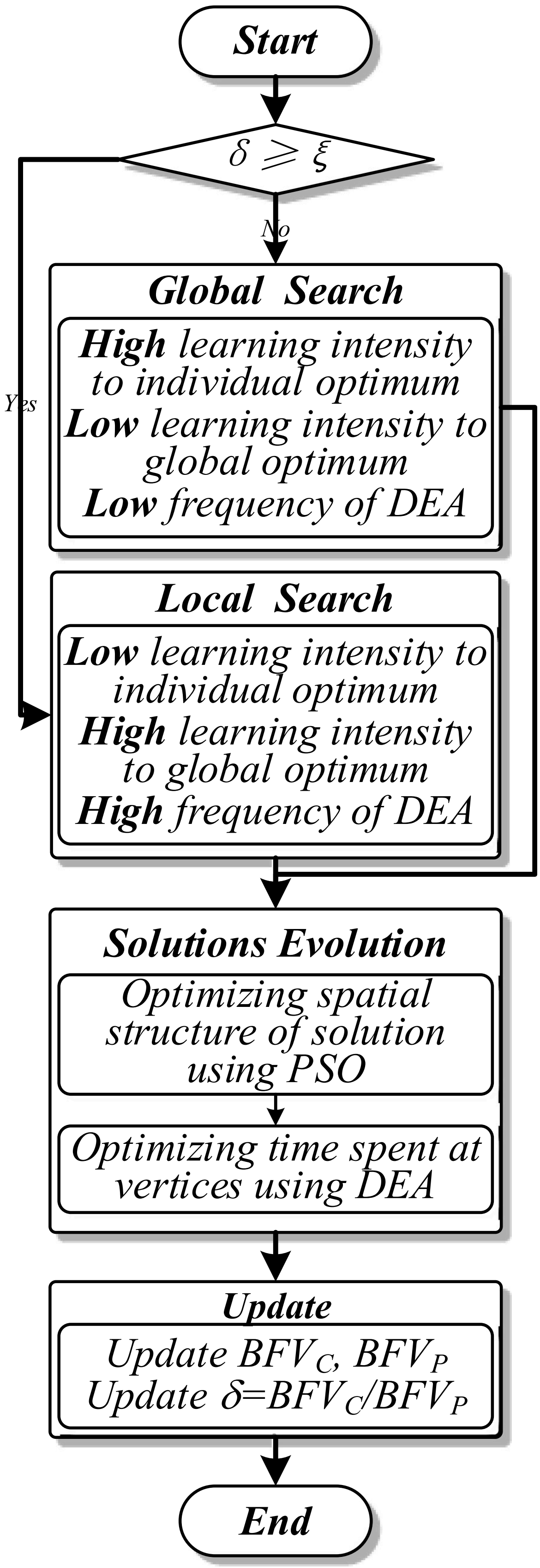

Fig. 3. Hybrid evolution procedure.

calculated according to Eq. (3.1), as listed in Column 3 of Table 3. The tourists were also required to record their budgeted time and must-visit/must-avoid POIs, as shown in the last two columns of Table 3. Lastly, a random selection of 50 tourists indicated that 23 were female and 12 were repeat tourists.

4.3. Algorithm parameters

In our system, the quality of itineraries is considerably influenced by the population size (P), number of iterations (G), learning factors of PSO (σ_1 and σ_2), the scale factor of DEA (Fd), and the crossover rate of DEA (Cr), respectively. Two critical parameters that considerably impact the PSO evolutionary performance are two learning factors (σ_1, σ_2). Previous studies have indicated that the two parameters are best set between 0.5 and 0.9 [60]. For DEA, the best values of Fd range from 0.1 to 0.3 to make a trade-off between performance and efficiency [86]. On the bases of previous analysis of and the situation in Kulangsu, our algorithm parameter settings are shown in Table 4.

4.4. System performance evaluation

We conducted two tests to verify the performance of our system. (1) By comparing with existing TTDP algorithms, we evaluate the utility of the itinerary designed using the heuristic approach (PSO-DEA) and the efficiency of algorithm operation. (2) We further evaluate the adaptability of our system by simulating six scenarios.

4.4.1. Evaluation of the heuristic approach

We designed itineraries using our approach and the four baseline methods: GA, PSO, ACO, combination of GA and DEA (GA-DEA), and combination of ACO and DEA (ACO-DEA). To avoid random errors, each method is used to design the itinerary for each tourist 30 times and takes the average utility of the 30 results. Paired sample t-tests were conducted to verify whether there are any significant differences among the utility achieved using the five methods. The results are shown in Tables 5 and 6. For the first pair (PSO-DEA–GA), the gap mean was 3.815, and PSO-DEA gained a significantly higher utility ($M = 43.657$, $SD = 6.528$) than GA ($M = 39.842$, $SD = 7.339$) ($t(50) = 12.591$, $p < 0.05$). Analogously, the second (PSO-DEA–PSO) and third (PSO-DEA–ACO) pairs indicated a clear advantage of our approach compared with PSO and ACO in improved utility. However, the test results for the fourth (PSO-DEA–GA-DEA) and fifth (PSO-DEA–ACO-DEA) pairs showed no significant difference.

We further evaluate the efficiency of the five approaches, i.e., the approaches' computational time. Tables 7 and 8 present the results of the paired t-tests. It is obvious that the computational time of GA, PSO, and ACO is superior to our approach (as shown in Pairs 6–8 in Table 8). However, given the significant shortcomings of these three approaches in terms of utility (as shown in Tables 5 and 6), while the goal of the system is to balance efficiency and performance. Therefore, we focus on evaluating the difference in computational time of Pair 9 (PSO-DEA–GA-DEA) and Pair 10 (PSO-DEA–ACO-DEA). For the ninth pair (PSO-DEA–GA-DEA), the computational time gap mean was -2.415, and the results indicated that the computational time of PSO-DEA was considerably smaller ($M = 8.052$, $SD = 1.746$) than that of GA-DEA ($M = 10.467$, $SD = 1.644$) ($t(50) = -7.916$, $p < 0.05$). Analogously, the tenth pair (PSO-DEA–ACO-DEA) also indicated the significant advantage of our approach compared with ACO-DEA in terms of improved efficiency. The results indicated our heuristic approach (PSO-DEA) is ideally situated for the problem owing to its high search efficiency [83,87]. From this perspective, although PSO-DEA, GA-DEA, and ACO-DEA achieved similar utility, our approach was clearly better in terms of computational efficiency.

4.4.2. Evaluation of TRS

This subsection further evaluates the adaptability of our system. To

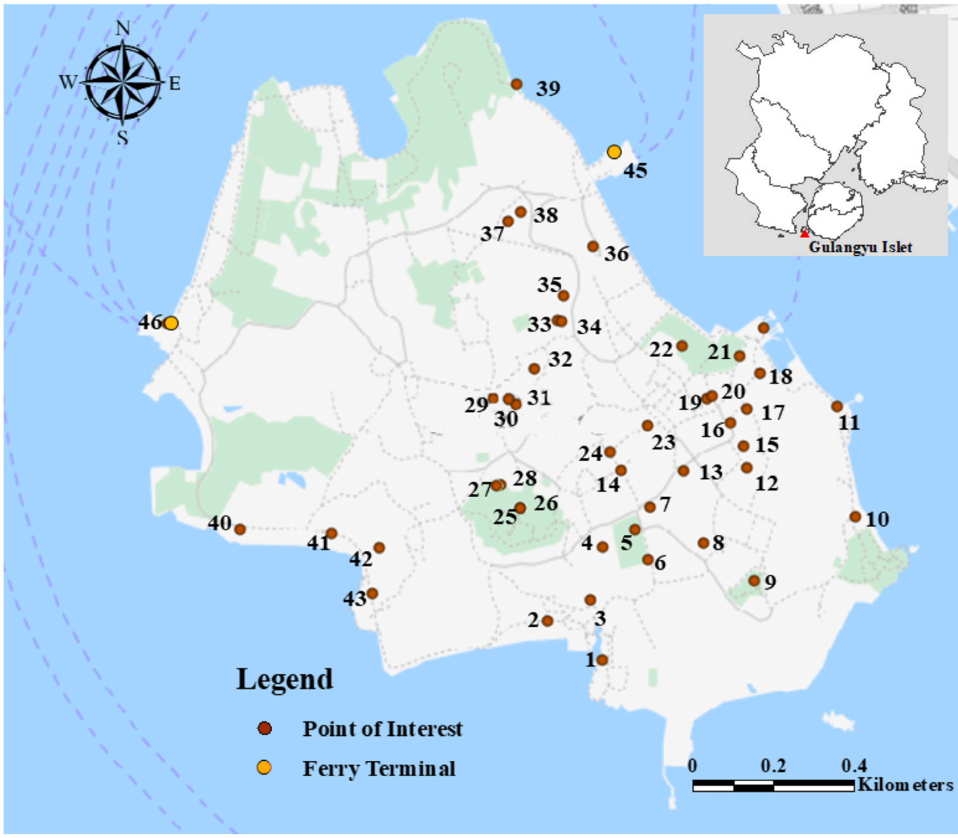

Fig. 4. Map of Kulangsu islet.

Table 2
Basic information of vertices in Kulangsu.

Type of vertices	Nos.	Names	Time windows	t_i (min)	POIs' feature vectors
POI	v_1	Shuzhuang Garden	[05:00–21:30]	60	[0.68, 0.75, 0.48, 0.30, 0.80, 0.78, 0.05, 0.80]
POI	v_2	Gangzaihou Seaside Resort	[00:00–24:00]	10	[0.40, 0.95, 0.23, 0.35, 0.35, 0.23, 0.00, 0.23]
….	….	….	….	….	….
POI	v_{44}	Kulangsu Piano Ferry Terminal	[00:00–24:00]	5	[0.25, 0.65, 0.25, 0.25, 0.48, 0.40, 0.00, 0.38]
Start-end locations	v_{45}	Sanqiutian Ferry Terminal	[00:00–24:00]	–	–
Start-end locations	v_{46}	Neicuoao Ferry Terminal	[07:20–18:40]	–	–

Table 3
Basic tourist information.

Tourists	Tourists' feature vector	Tourists' preference vector	Budgeted time	Compulsory POIs	Must-avoid POIs
1	[0.85, 0.95, 0.80, 0.85, 0.50, 0.40, 0.10, 0.20]	[0.84, 0.91, …, 0.87]	7 h [10:00–17:00]	28, 35	None
2	[0.35, 0.10, 0.25, 0.30, 0.95, 0.90, 0.80, 1.00]	[0.82, 0.51, …, 0.73]	6 h [10:00–16:00]	25	None
…	…	…	…	…	…
50	[0.85, 0.80, 0.65, 0.60, 0.80, 0.65, 0.75, 0.70]	[0.92, 0.82, …, 0.89]	9 h [9:00–18:00]	None	None

better simulate various changes in reality, we simulate six scenarios: (1) temporarily cancel visiting a planned POI, (2) temporarily add an unplanned POI, (3) stay longer than planned in a POI, (4) stay shorter than planned in a POI, (5) spend more time in traffic than expected, and (6) spend less time in traffic than expected. For comparison, a static system is used as a baseline. When the actual itineraries of tourists deviate from the planned ones, the static system will not redesign new itineraries. Therefore, tourists can only refer to the original itinerary to complete the rest of the trip. That is, when tourists temporarily visit an unplanned POI or spend more time on POIs or traffic than expected, these changes will prevent tourists from visiting some planned POIs owing to time constraints. By contrast, when tourists temporarily cancel visiting a planned POI or spend less time on POIs or traffic than expected, these changes will cause them to end their trip ahead of schedule, thereby achieving less utility than expected.

First, we design an itinerary for each tourist, as shown in Table 3, according to the basic information of tourists and POIs. Second, we randomly select scenarios from the six scenarios to simulate real situations. Third, our system will adjust the itinerary for each tourist according to the changes, whereas a static system will not redesign new itineraries. Thus, tourists can only refer to the original itinerary to complete the rest of the trip. Fig. 5 shows the three utility values for each

Table 4
Algorithm parameters.

Parameter	G	P	Global search				Local search			
			Improve PSO		DEA		Improve PSO		DEA	
			σ_1-G	σ_2-G	Fd-G	Cr-G	σ_1-L	σ_2-L	Fd-L	Cr-L
Value	200	50	0.9	0.5	0.1	0.5	0.5	0.9	0.3	0.5

Table 5
Paired sample statistics.

		Mean	N	Std. Deviation	Std. Error Mean
Pair 1	PSO-DEA	43.657	50	6.528	.923
	GA	39.842	50	7.339	1.038
Pair 2	PSO-DEA	43.657	50	6.528	.923
	PSO	38.126	50	8.962	1.267
Pair 3	PSO-DEA	43.657	50	6.528	.923
	ACO	40.662	50	6.777	.958
Pair 4	PSO-DEA	43.657	50	6.528	.923
	GA-DEA	43.772	50	7.066	.999
Pair 5	PSO-DEA	43.657	50	6.528	.923
	ACO-DEA	43.887	50	6.617	.936

Table 6
Paired sample test.

		Paired Differences					t	df	Sig. (2-tailed)
		Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference				
					Lower	Upper			
Pair 1	PSO-DEA–GA	3.815	2.142	.303	3.206	4.424	12.591	49	.000***
Pair 2	PSO-DEA–PSO	5.531	3.843	.544	4.439	6.624	10.177	49	.000***
Pair 3	PSO-DEA–ACO	2.996	1.205	.170	2.653	3.338	17.572	49	.000***
Pair 4	PSO-DEA–GA-DEA	−.115	1.436	.203	−.523	.294	−.564	49	.575
Pair 5	PSO-DEA–ACO-DEA	−.230	1.217	.172	−.576	.116	−1.334	49	.188

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

Table 7
Paired sample statistics (Computational time).

		Mean	N	Std. Deviation	Std. Error Mean
Pair 6	PSO-DEA	8.052	50	1.746	.247
	GA	6.645	50	1.991	.282
Pair 7	PSO-DEA	8.052	50	1.746	.247
	PSO	6.137	50	1.948	.276
Pair 8	PSO-DEA	8.052	50	1.746	.247
	ACO	7.962	50	1.732	.245
Pair 9	PSO-DEA	8.052	50	1.746	.247
	GA-DEA	10.467	50	1.644	.233
Pair 10	PSO-DEA	8.052	50	1.746	.247
	ACO-DEA	11.538	50	1.727	.244

tourist, corresponding to the itineraries designed by the two systems (red and green lines represent the adaptive and static systems, respectively). For comparison, we introduce a benchmark, which denotes that tourists' travel in strict accordance with planned itineraries (as denoted by the blue lines). That is, the environment has not changed and tourists have not changed their itinerary, which is an ideal situation. As shown in Fig. 5, the utility value obtained by the benchmark method is significantly higher than that obtained by the other two systems. This phenomenon indicates that readjusting the itineraries during the trip will relatively reduce tourists' utility. We further analyze the performance of the adaptive and static systems through paired sample t-tests to identify which system obtained greater utility and smaller decline than the benchmark.

The mean and standard of the utility estimated by the two systems are presented in Table 9 (Pair 1). Meanwhile, the mean and standard of the differences between the benchmark are shown in Table 9 (Pair 2).

The results of the paired sample t-test are shown in Table 10. The gap mean was 3.409 for the first pair, indicating that our system obtained significantly remarkable utility ($M = 39.854$, $SD = 4.161$) compared with the static system ($M = 36.445$, $SD = 4.135$; $t(50) = 13.947$, $p < 0.05$). For the second pair, the gap mean was − 1.073, implying that the adaptive system achieved a markedly smaller decline ($M = 2.336$, $SD = 1.421$) than the static system ($M = 3.409$, $SD = 1.729$; $t(50) = -3.177$, $p < 0.05$). The results indicate that our system is beneficial in improving the adaptability of the designed itineraries.

4.5. Discussion

The statistical results shown in Section 4.3 indicate that our adaptive system achieved significantly better performance than the static system. To better explain why our system can provide more adaptable itineraries for tourists, we take the first two tourists in Table 3 as examples.

The budgeted time for the first tourist was 7 h (10:00 to 17:00), and his must-visit POIs are v_{28} (Xilin Villa) and v_{35} (Organ Museum). According to the information related to tourists and POIs, we designed the itinerary for this tourist, denoted as "benchmark": v_{45}–v_{36}–v_{22}–v_{21}–v_{11}–v_{10}–v_9…–v_{28}–v_{35}–v_{45} (shown as blue lines in Fig. 6). The utility of this itinerary was 48.492. However, after visiting v_{10}, instead of visiting v_9 as planned, the tourist temporarily visited v_{12} (Hai Tian Tang Gou Mansion) and spent 60 min at v_{12}. In response to this temporary change, our system replans the rest of the itinerary for this tourist: v_{12}–v_{13}–v_{14}–v_{24}–v_{25}–v_{26}–v_{27}–v_{28}–v_{35}–v_{45}, denoted as "adaptive" (shown as red lines in Fig. 6). The utility of this itinerary was 45.748, which is slightly lower than the benchmark. By contrast, the static system did not readjust the tourists' rest trip. Thus, after visiting the unplanned POI (v_{12}), the tourist still followed the designed trip. However, visiting v_{12} increases the time of the planned itinerary, such that the last few POIs of the planned itinerary cannot be visited because of time budget, which is a hard constraint. After visiting v_{12}, the tourist particularly followed the designed trip and did not have sufficient time to visit v_{27}, v_{28}, and v_{35}. The rest trip after v_{12} is v_{12}–v_9–v_8–v_6–v_4–v_3–v_1–v_2–v_{25}–v_{45}, denoted as "static" (shown as violet lines in Fig. 6). The utility of this itinerary was 41.836, which is significantly lower than the benchmark and adaptive system. Moreover, v_{28} and v_{35}, two POIs that the tourist must visit, were not included in the actual itinerary, thereby affecting the utility and also making the itinerary potentially unacceptable to the tourist [88].

For the second tourist in Table 3, her budgeted time was 6 h (10:00 to 16:00) and her must-visit POI is v_{25} (Sunlight Rock). According to the

Table 8
Paired sample test (Computational time).

		Paired Differences					t	df	Sig. (2-tailed)
		Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference				
					Lower	Upper			
Pair 6	PSO-DEA–GA	1.407	.792	.112	1.182	1.632	12.560	49	.000***
Pair 7	PSO-DEA–PSO	1.914	1.317	.186	1.54014	2.289	10.279	49	.000***
Pair 8	PSO-DEA–ACO	.089	.243	.034	.02017	.158	2.597	49	.012*
Pair 9	PSO-DEA–GA-DEA	−2.415	2.158	.305	−3.029	−1.802	−7.916	49	.000***
Pair 10	PSO-DEA–ACO-DEA	−3.486	2.504	.354	−4.198	−2.775	−9.845	49	.000***

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

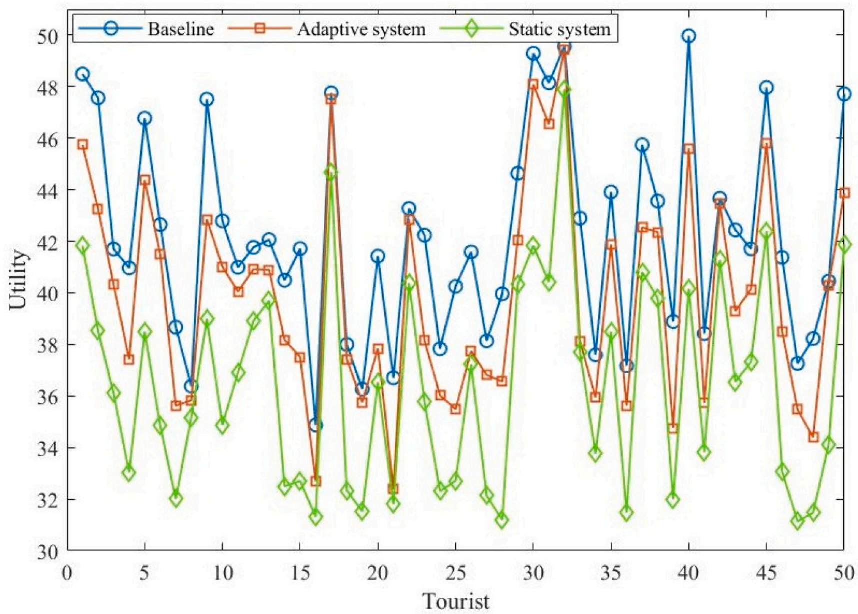

Fig. 5. Average utility for each tourist (benchmark and two systems).

information related to tourists and POIs, we designed the itinerary for this tourist, denoted as "benchmark": v_{46}–v_{40}–v_{41}–v_{42}…–v_{38}–v_{37}–v_{45} (shown as blue lines in Fig. 7). The utility of this itinerary was 47.563. However, after visiting v_{41}, she temporarily canceled visiting v_{42} (Shell Museum), which is a planned POI. In response to this temporary change, our system replanned the rest of the itinerary for this tourist as v_{41}–v_{25}–v_{27}–v_{28}–v_{30}…–v_{37}–v_{39}–v_{45}, denoted as "adaptive" (shown as red lines in Fig. 7). The utility of this itinerary was 43.260, which is slightly lower than the benchmark. By contrast, the static system did not readjust the tourists' rest trip. Thus, given the cancellation of the scheduled visit of v_{42}, the tourist continued the rest of the original itinerary. However, the cancellation of v_{42} reduced the time of the planned itinerary, such that the tourist completed the entire trip earlier than planned, resulting in less utility (i.e., 38.527, which is significantly lower than the benchmark and the adaptive system). The actual itinerary of the tourist is denoted as "Static," as shown by the violet lines in Fig. 7.

5. Conclusions

At present, the tourism market is controlled by the demand of tourists for personalized experiences [1,2]. As such, tourists' understanding of destinations' social conditions and culture is facilitated by personalized TRS, which significantly contributes to the enhancement of their experiences and destinations' tourism competitiveness [5,35]. However, the uncertainty of the external environment and tourists' changing on-site behavior have brought immense challenges to TRS. Therefore, this study proposed an adaptive TRS to adapt to uncertain environments and the changing behavior of tourists. Compared with existing TRSs, our system is more superior because it contains the mainstream functions of TRS and improves system operation efficiency in various ways to ensure real-time response. The results of the case study verify the superiority of the adaptive system over existing systems, especially in the face of uncertain environments and changing behavior of tourists. Our adaptive TRS may be of considerable interest to tourism practitioners because of the development of customized experiences that are currently prevailing in the tourism market.

This study has methodological and practical contributions for tourism research. Methodologically, we designed a novel system to provide tourists with adaptive and personalized itineraries. This system fully considers the uncertainties during trips. An innovative approach of tourist profile learning is adopted to accurately collect tourist preferences. Moreover, a hybrid optimization structure is adopted to balance the efficiency and performance of itinerary design. Lastly, a dynamic mechanism is proposed to enhance the flexibility and adaptability of the proposed system.

Practically, utilizing the latest AI technologies to provide

Table 9
Paired sample statistics.

	System	Mean	N	Standard deviation	Standard error mean
Pair 1	Adaptive	39.854	50	4.161	0.589
	Static	36.445	50	4.135	0.585
Pair 2	Adaptive	2.336	50	1.421	0.201
	Static	3.409	50	1.729	0.244

Table 10
Paired sample tests.

		Paired differences					t	df	Sig. (2-tailed)
		Mean	Standard deviation	Standard error mean	95% Confidence interval of the difference				
					Lower	Upper			
Pair 1	Adaptive–Static	3.409	1.729	0.244	2.918	3.90053	13.947	49	0.000***
Pair 2	Adaptive–Static	−1.073	2.389	0.338	−1.752	−0.39424	−3.177	49	0.003**

*$p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

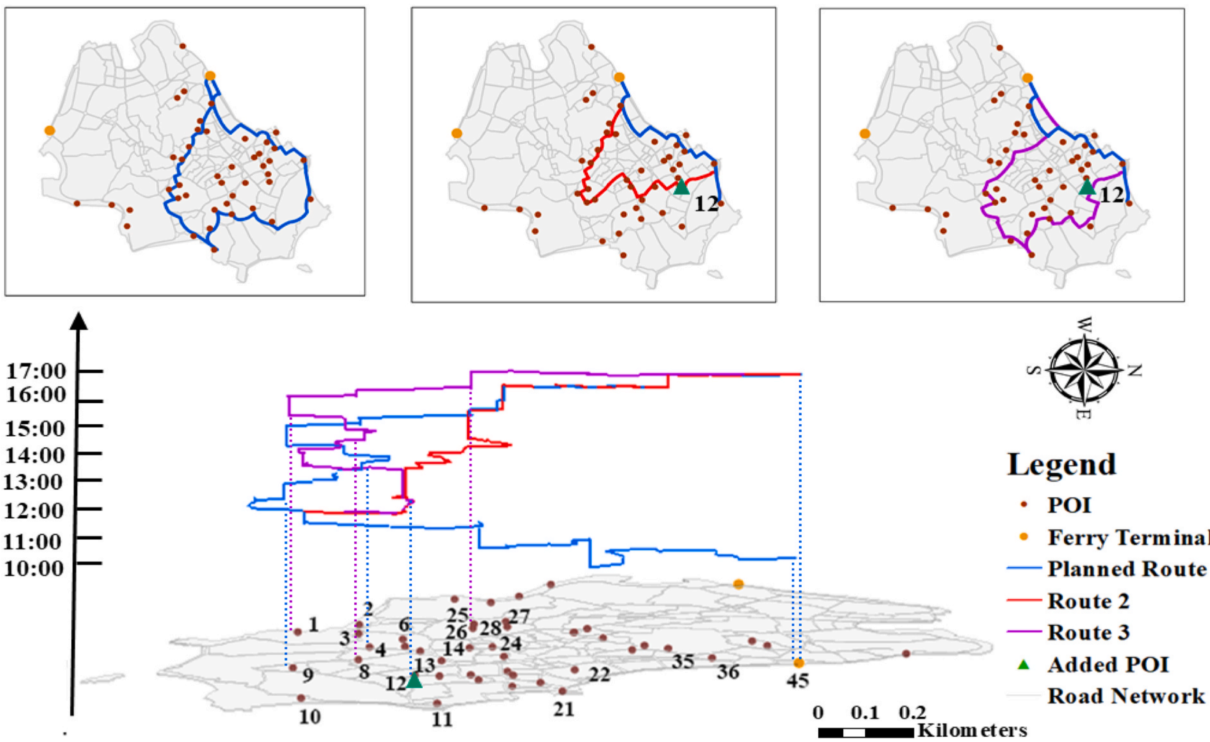

Fig. 6. Three itineraries for Tourist 1.

personalized services for tourists will significantly improve their tourism experience, satisfaction, and loyalty, thereby bringing better development opportunities for tourism enterprises [89]. Our adaptive system further improves the intelligence level of TRS, which is intended to provide tourists with real-time and personalized services in uncertain and changing contexts. Our system can be integrated by tourism destination management organizations in upgrading their information service platforms as part of their smart tourism construction, which will enhance the competitiveness of destinations [90].

We also propose worthy avenues for future exploration. First, for adaptive recommendation systems, real-time response speed is extremely important, which presents higher requirements for the operation efficiency of these systems. Therefore, future research to further improve the efficiency of the algorithm is extremely necessary and valuable. Second, for island tourism destinations with multiple ferry terminals, the choices of ferry terminals should be considered when planning trips for tourists. That is, choosing different terminals implies different locations and also different arrival and departure sailings. Undoubtedly, considering the choices of terminals can affect the tourist route structure, thereby increasing the complexity of the problem. Lastly, the continuous impact of the COVID-19 pandemic make tourists tend to avoid overcrowded places. Thus, collecting the number of tourists at POIs in real time and combining this information into TRS have immense potential and value.

CRediT authorship contribution statement

Yuguo Yuan: Conceptualization, Methodology, Software, Investigation, Formal analysis, Data curation, Writing – original draft. **Weimin Zheng**: Conceptualization, Methodology, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

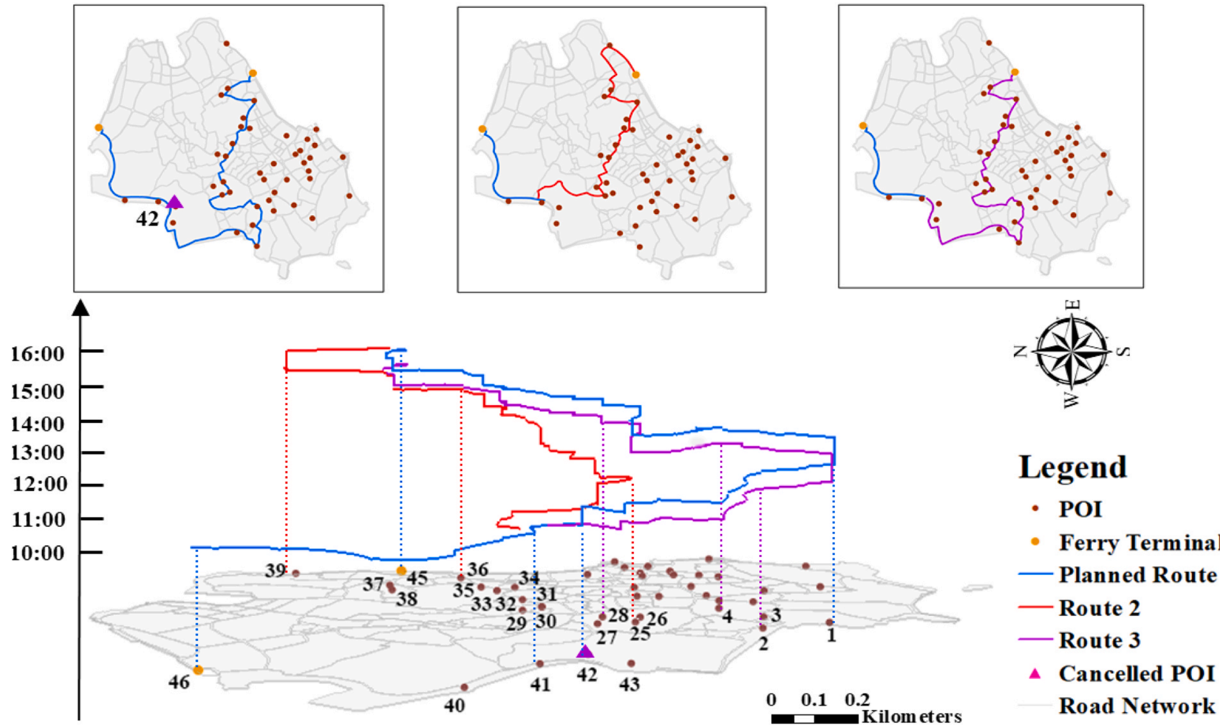

Fig. 7. Three itineraries for Tourist 2.

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